

Day 1: Cybersecurity Fundamentals Worksheet

Name: V Ananya Date: 14/06/26

Part 1: The CIA Triad - Real World Examples

For each scenario below, identify which CIA principle (Confidentiality, Integrity, Availability) is most affected:

Scenario	CIA Principle	Explanation (1-2 sentences)
Someone reads your private messages without permission	Confidentiality	Reading private messages is a breach of confidentiality.
A hacker modifies your bank balance	Integrity	Changing the balance challenges the integrity of a hacker
A DDoS attack makes a website unreachable	Availability	Makes the website unavailable due to ddos attack
Your password is stolen and used to login	Confidentiality	Account can be accessed by anyone which is a threat to personal information

Part 2: AAA Security Model

Match the term:

- Verifying who you claim to be : Authentication
- Checking if you have permission to access a resource: Authorization
- Logging what actions you perform: Accounting

a) Checking if you have permission to access a resource b) Verifying who you claim to be c) Logging what actions you perform

Part 3: Types of Attackers

Draw lines to match:

1. Script Kiddie → 1) Uses tools made by others, little skill
 2. Hactivist → 2) Motivated by political/social causes
 3. Insider → 3) Employee or trusted person
 4. State-Sponsored → 4) Government-backed hackers
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Part 4: Threat vs Vulnerability vs Risk

Fill in the blanks:

- A **vulnerability** is a weakness in a system (e.g., unpatched software)
 - A **threat** is what could happen (e.g., data theft)
 - A **risk** is the likelihood and impact of a threat exploiting a vulnerability
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Part 5: Security Domains - Spot the Domain

Identify which security domain applies:

Situation	Domain
Setting up firewall rules	Network
Installing antivirus on laptop	Host
Writing secure password code	Application
Encrypting customer database	Data
Situation	Domain

Lock on server room door	Physical
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Part 6: Real Breach Analysis

Pick ONE real breach from the table and describe what went wrong:

Breach	Year	What Happened
Equifax	2017	Unpatched vulnerability exposed millions of people's data and confidential information
SolarWinds	2020	A group of hackers installed a backdoor to the system's software slowly escaping the security checks, the software update was installed by many people which led to their systems getting hacked
Colonial Pipeline	2021	Ransomware attack that happened on a fuel company which blocked the pipelines for fuel this even led to shortage of fuel for a while where 5 million dollars in bitcoin was paid
Twitter Bitcoin Scam	2020	Hijacked high profile blue check accounts to send a fake bitcoin link and generated over 180 million dollars before shutting down the posting abilities for blue check marks

Your chosen breach: Colonial Pipeline

Root cause (vulnerability): Easy password access, no rotation of credentials or 2 factor authentication

What CIA principle was violated: Availability, those pipelines which got hacked nearly stopped supplying fuel for a while, to the point a national crises on fuel was announced

How it could have been prevented: By having 2FA, changing of credentials in a timely manner.

Part 7: Responsible Disclosure

Scenario: You found a login bypass on your university's website.

What should you do? (List 3 steps)

1. Do not exploit or misuse the vulnerability.
2. Report it to the university's IT/security team.

3. Provide details and wait for them to fix it before public disclosure.

Part 8: Quick Reflection

What's one cybersecurity concept you want to learn more about? Why?

Finding out vulnerabilities in the software beforehand so I can tackle them and patch before it reaches the human hands where it can lead to exploitation

Submission

Submit this completed worksheet as `day1-worksheet.pdf`

Need help: Take a photo or scan - no special format required!